

Table of Contents

iMotions Lab > Getting Started	
Using the iMotions Study Builder	2

Using the iMotions Study Builder

This article explains how to create a new study in iMotions - including how to decide on a study type, configuring the different recording types, and how to build a screen-based experiment.

Creating a study is typically the first thing you do in iMotions. The steps to do so differ between studies involving a stimulus presentation on screen, and studies that only record data for example from cameras, screen grabs, eye-tracking glasses, or VR headsets. From iMotions 10 and onward, you can choose a study type and then be guided through the relevant configuration steps.

For studies involving a stimulus presentation, you can in the end define what your respondent will be presented with during the study, and define the order of stimuli. It is also possible to make variants of your experiment by generating additional study flows.

Getting started video

Let our Product Director, Kerstin Wolf, walk you through how to set up studies in iMotions and how to use the different study types (videos recorded in December 2023).

[Click here to download a video](#)

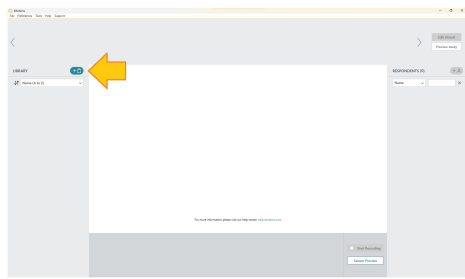
Study types video

[Click here to download a video](#)

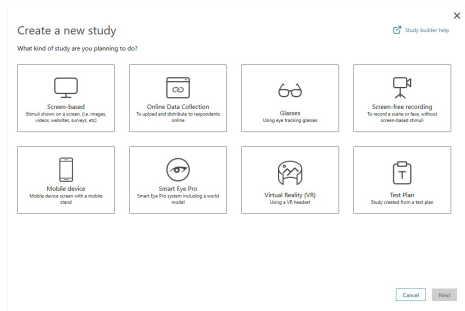
Making a study in iMotions

Adding a new study

A new study is added to the library by clicking on the "Add New Study" button or hitting Ctrl + N.



Please choose the study type you want to set up.



All study types available on a full license

Overview of study types

It is possible that you do not see all these study types because some of them require you have purchased the respective module

- Screen-based:** Choose this to create an experiment delivered on a computer screen in a lab. Your study can contain images, videos, audio files, surveys, or recordings of the interaction with websites or other applications on the PC. *Follow this article for further steps on how to configure a screen-based study.*
- Remote data collection** (previously known as online data collection, or ODC): Choose this to create an experiment that can be uploaded to my.imotions.com and shared with respondents via a link. Your study can contain images, videos, audio files, surveys, or record the interaction with websites. Data will be collected through the webcam.
- Glasses:** Choose this if you want to collect data from eye-tracking glasses, or have already recorded data from eye-tracking glasses that you want to import into iMotions.
 - Glasses live recording:** Records data from eye-tracking glasses connected to your PC. Data collection is initiated from iMotions, and therefore automatically synchronized with other biosensors, cameras, or live markers.
 - Sensor Sync for Glasses:** If data from eye-tracking glasses must be collected in an external application, it is possible to synchronize them with other biosensors, cameras, or live markers by using the sensor sync. Use this study type to record all data apart from the eye-tracking glasses.
 - Import glasses data:** Import eye-tracking glasses data collected outside iMotions for further analysis or synchronization with other sensors.
- Screen-free recording:** Choose this if your study does not require delivering a stimulus to respondents while collecting sensor data, for example in an interview situation where you are recording through cameras.
 - Scene recording:** Collect data from a scene camera to film what the respondent is doing during an experiment. For eye-tracking studies, you can combine this camera video with the eye-tracking data to measure their gaze on physical objects or screen grabs from other applications through a framegrabber.
 - Face recording:** Collect real-time data from the respondent camera synchronized with other biosensors, cameras, or live markers. To record face data with this study type, it is not required to add any screen-based stimuli to your study.
 - Import face recordings:** Import face videos collected outside iMotions for further analysis. *Learn how to initiate the processing and select what face to process.*
 - Import voice recordings:** Import voice recordings collected outside iMotions for further analysis. *Learn more about voice analysis here.*
- Mobile device:** Choose this if you want to record your respondents while they interact with apps on a mobile phone. This requires that you can get a real-time recording of the phone screen e.g. through a frame grabber. *Learn more about how to configure your screen-based eye tracker for data collection on a phone or tablet testing using a mobile device stand here.*
- Smart Eye Pro:** Choose this if you want to record data from your Smart Eye Pro system
 - SE Pro + Scene cam:** Collect data using your Smart Eye Pro system plus an additional external scene camera. This will require world model scene calibration step. Data will be mapped from the world model into the scene camera view, but only gaze data from the world model will be mapped.
 - SE Pro only:** Collect data using your Smart Eye Pro system, using only the world model, with no additional scene cameras. Data will be mapped into a simple grid view of the world model, and only gaze data from the world model will be mapped.
 - SE Pro + Gaze in scene:** Collect data using your Smart Eye Pro system plus an external scene camera with gaze calibration. Using a world model is optional. This will require an additional scene camera calibration. Data will be mapped into the scene camera view.
 - Import Smart Recorder :** Import eye tracking and scene camera data collected in the Smart Eye Smart Recorder for further analysis
- Virtual Reality (VR):** Choose this if you want to record or import data from a VR headset
 - Unity :** Record VR experiences that have been designed in Unity.
 - 360 stimuli:** Create a study containing 360-degree images or videos to be presented to your respondents in a defined order on a head-mounted display.
 - Sensor Sync for VR:** If eye-tracking data from VR must be collected in an external application, it is possible to synchronize them with other biosensors, cameras, or live markers by using the sensor sync. Use this study type to record all data apart from the eye-tracking data.
 - Import VR recordings:** Import VR experiences collected outside iMotions for further analysis. Imported VR data can be merged with an existing sensor sync for VR study to manually synchronize VR eye-tracking data with recordings from other biosensors, cameras, or live markers.
- Test Plan:** Choose this to create an experiment delivered on a computer screen in a lab. Other than the option "Screen-based", this allows you to define the stimulus sequence through a CSV file rather than building it in the user interface (UI).

Configuration of the screen size

This is only relevant for the screen-based study type

If you choose to set up a screen-based experiment, iMotions will ask you about the resolution and physical size of the screen you will use to display stimuli to the respondent.

This is important so that iMotions can correctly display the stimuli and calculate eye-tracking statistics for your setup.

Automatic screen detection

iMotions will look up the available PC screens and attempt to read out their width, height, and resolution.

The specs of the screen that is configured under Global Settings > Slide Show > Screen will be used as a default.

List of connected screens detected by iMotions

Please note:

1. This automated way of accessing screen parameters is not always possible. We, therefore, recommend to always double-check these values.
2. Only the primary and secondary screens (as configured in the Windows Display Settings) can be configured in Global Settings. In case you have more than two screens in use with your PC, please ensure that they are set to primary and secondary screens to be used for stimulus presentation.
3. The screen resolution used by iMotions takes into account if a screen is configured to scale. As an example: A screen with 1920 × 1080 px resolution scaling at 125% will be discovered as having a 1536 × 864 px resolution.

Manual measurement

To have full control over the screen parameters, we recommend measuring the width and height of your screen with a measuring tape (in centimeters) and entering the values.

Then enter the resolution of the screen (in pixels).

We recommend that you save this custom screen by entering a name for it. Saved custom screens can be picked in future studies.

Enter measured screen parameters and save them as a custom screen

Saved parameters can be selected from the dropdown

Select sensors

Select the sensors you will be using to collect data for your study.

If a sensor is currently enabled in Global Settings, iMotions will auto-select this sensor.

Select the sensors you want to use in your study

Name your study

Finally, give your study a name.

Adding extra notes is optional.

Enter a study name

You can always rename your study afterward.

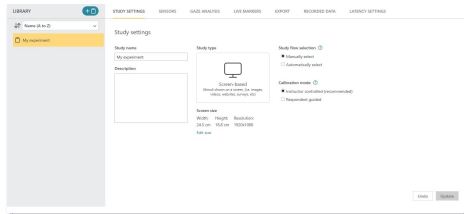
Configuring a study

You can review all settings defined in the previous steps under Study settings, and the selection of your sensors under Sensors on the study level.

Study settings: Screen-based study

Under *Study settings*, you can

- Review and edit your study name as well as description and
- Review and edit the defined screen parameters.
- If you are running a between-subject design (A/B test) with more than one study flow, you can select how a respondent is allocated to a study flow. The experimenter can manually select a flow at the start of a recording, or a respondent can be randomized into a flow automatically selected by iMotions.
- If you are running a screen-based test with eye tracking, you can select the *calibration mode* to choose whether you want the experimenter to guide the respondent through the setup steps (recommended) or if you want iMotions to guide them.



Study settings for a screen-based study with multiple flows and eye tracking enabled

“Screen” configuration for other study types

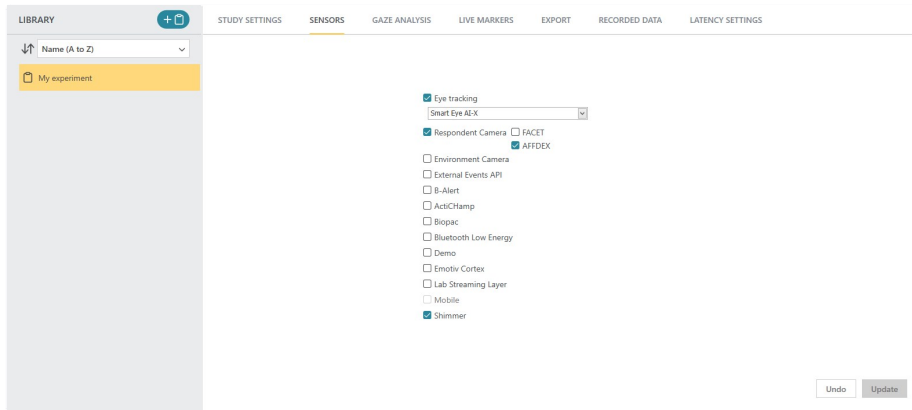
If a study is of a different type than “screen-based”, you do not have to configure the width, height, and resolution of a respondent screen when setting up your study.

However, iMotions needs information about the resolution and physical dimensions of the study material to ensure correct calculations of eye-tracking metrics.

- **Remote data collection:** A standard resolution of 1920 x 1080, as well as a standard screen with a 47.4 cm width, 29.6 cm height, and 65 cm distance, is pre-populated, but not accessible to the user for edits.
- **Glasses live recording:** Based on the selected device, the resolution of the device's world camera is automatically filled in. A visual area size is calculated based on this world camera's field of view as specified by the manufacturer. The visual area size is based on a plane at a 100 cm distance.
- **Sensor Sync for Glasses:** The resolution of the screen configured under Global Settings > Slide show > Screen is automatically used for the study. If the study is executed on a PC with a different resolution, it is recommended to update the study settings to ensure a correct display of the synchronization stimulus.
- **Import glasses data:** Based on the selected device, the resolution of the device's world camera is automatically filled in. A visual area size is calculated based on this world camera's field of view as specified by the manufacturer. The visual area size is based on a plane at a 100 cm distance. *Please note: These values are not available for all supported devices. If they are unavailable, iMotions defaults to the parameters of the screen currently selected under Global Settings > Slide show > Screen*
- **Scene recording, Face recording, Voice recording, Mobile device:** The resolution of the screen currently selected under Global Settings > Slide show > Screen is used by default, along with a 47.4 cm width and 29.6 cm height of an assumed standard screen. *Please note: If data is recorded from a frame grabber or camera with a resolution different from the study resolution, we recommend that you change the study resolution to be aligned with the video resolution. If an eye tracker is used, the study resolution will be used together with the visual area size to calculate gaze angles (see e.g. this article for more information). Hence, it is important to replace the standard values for the visual area with manually measured values. In a mobile device study, for example, this would be the area of the phone's active display. If the study resolution does not match the resolution of the phone display's recording from the frame grabber, iMotions inserts a black frame around the recording of the active display. The visual area then consists of the display plus the frame.*
- **Smart Eye Pro:** No resolution, width, and height given in the study settings. Imported Smart Recorder data will be set to the resolution of the imported video. *If parameters are unavailable, iMotions defaults to the parameters of the screen currently selected under Global Settings > Slide show > Screen*
- **VR:** Since iMotions will record the VR experience through a screen recording of the application, the resolution of the screen currently selected under Global Settings > Slide show > Screen is used by default, along with a visual area size based on the device's field of view specs and a plane at 100 cm distance.

Sensors

If you need to change the selection of sensors for your study, you can click the checkboxes under Sensors.

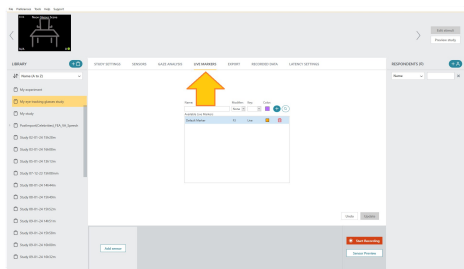


Sensor selection in the main app

Live markers

If you want to add live markers to your study, you can do so on the study level.

Please follow this article for a description of how to work with live markers.



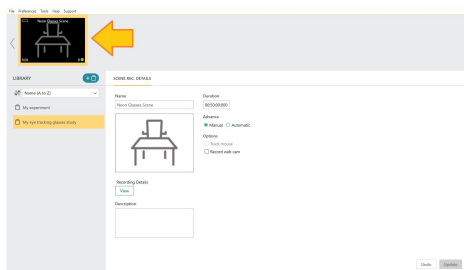
The UI to add live markers

Recording properties

If you are creating a screen-based study, this section is irrelevant for you. Jump ahead to “Designing an experiment” instead.

Study types that mainly consist of a recording without any stimulus delivery logic, for example, studies with eye tracking glasses or mobile phone testing, are straightforward to set up by selecting the respective study type.

If you need to configure your recording, for example defining a maximum recording time, you can do so by clicking on the stimulus and changing its properties.



Click on the stimulus to review its properties

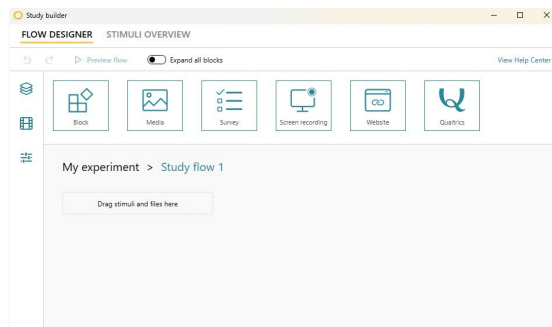
Designing a study flow

For studies that involve a presentation of stimuli, you can click “Edit stimuli” to open the study builder.

Here, you can import different kinds of stimuli and configure the study flow, i.e. when and for how long the stimuli of your experiment will be presented to the respondent.

What is in the UI?

- **Flow designer (tab):** Allows you to add and edit stimuli as well as to decide on the order in which they will be delivered during the experiment
 - **Study flows (menu option):** Here, you can manage your study flows, i.e. different versions of your experiment
 - **Repository (menu option):** Here, you can see what stimuli are being used in the current study flow as well as all other stimuli you added to your study but not using in this flow. This helps you to manage stimuli when designing multiple study flows.
 - **Properties (menu option):** Allows you to change the properties of one or many stimulus/ stimuli or block(s), such as name, presentation duration, display order et cetera.
- **Stimuli overview (tab):** An overview of all stimuli used in the study with their properties. Allows to multi select and batch edit stimulus properties
 - **Properties (menu option):** Allows you to change the properties of one or many stimulus/ stimuli, such as name, presentation duration, et cetera.



The available menu options

Stimulus options

Stimuli can be added by clicking on the stimulus type icons at the top and dragging them to their dedicated location.

For Media stimuli, it is also possible to open the Windows file explorer and drag in multiple stimuli at the same time.

1. **Block:** Allows you to group stimuli together for complex study design
2. **Media:** Images (in JPEG, PNG, BMP format), videos (in MP4 or WAV format), and audio (in MP3 or WAV format, can be combined with an image file) files
3. **Survey:** For quantitative (single/ multiple choices, Likert scales, and key press) and qualitative (open text) survey responses
4. **Screen recording:** Records the PC screen while respondents interact with an application.
5. **Website:** Records the PC screen while respondents interact with the indicated website.
6. **Scene camera:** Records the video from a webcam that was configured as Scene Camera in Global Settings
7. **360 Image (VR):** Presents a 360-degree image in the VR headset
8. **360 Video (VR):** Presents a 360-degree video in the VR headset
9. **Qualtrics:** Opens the link to a survey set up in Qualtrics
10. **Face recording:** Records the video from a webcam that was configured as Respondent Camera in Global Settings without concurrent stimulus presentation

Depending on the study type, you have different stimuli available.

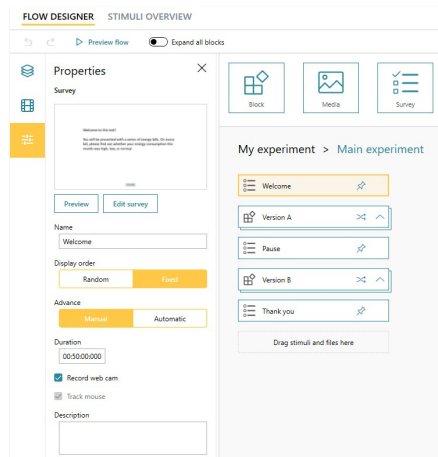
	Screen-based	Remote data collection	Screen free recording	Mobile device	VR
Media	x	x	x	x	
Survey	x	x	x	x	x
Screen recording	x		x	x	x
Website	x	x	x	x	
Scene camera			x	x	
360 stimuli (VR)					x
Qualtrics	x	x	x	x	
Face recording			x		

Editing stimulus properties

Drag and drop stimuli in the *flow designer* to change their order in the flow. Alternatively, you can activate randomization.

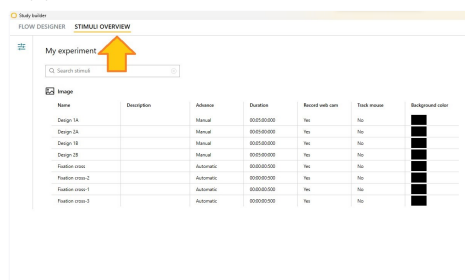
- The pin icon means that this stimulus has a fixed position in the study flow
- The arrow icon means that this stimulus is randomized

Stimulus properties can be edited in the left-side menu after selecting the stimulus.



The properties menu

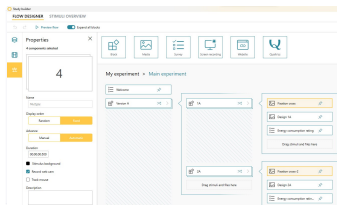
Stimulus properties can be reviewed in the *stimuli overview*



The stimuli overview

Multiselection

It is possible to multiselect stimuli and edit those properties that are shared between them, such as their presentation duration.

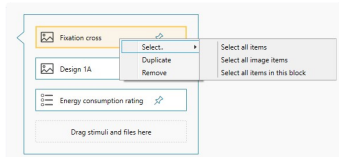


Multiselect stimuli to batch edit their properties

Beyond ctrl-clicking and drag-selecting stimuli with the keyboard and mouse, while working in the flow designer, it is possible to multiselect stimuli through the right-click context menu.

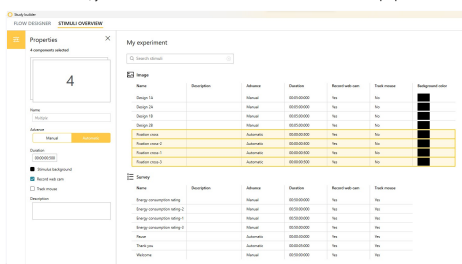
This allows you to select

- All items
- All items of the same type (e.g. all image stimuli, all website stimuli, all survey slides, etc.)
- All items in this block (Please note that if you click on a block, this will select this block and other items on the same hierarchical level and not all items in the block you clicked on.)



Multiselect options in the right click menu

In the **stimuli overview**, you can multi select similar stimuli and batch edit their shared properties. The search bar allows you to search for specific stimulus names.



Multiselection in the stimuli overview

Duplication

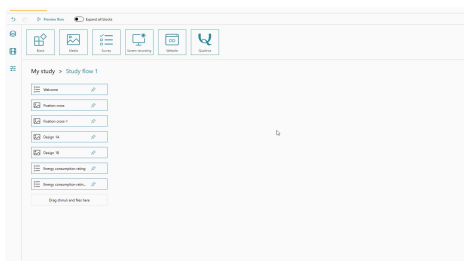
You can duplicate a single stimulus, a multiselection of stimuli, or entire blocks (with the items inside it) through the right-click menu or Ctrl+D in the flow designer

Creating blocks

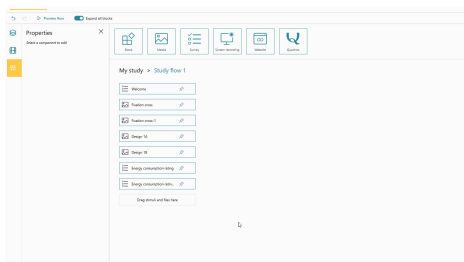
Blocks allow you to group stimuli and build a more complex experiment structure, making sure that certain stimuli always follow after each other.

You have different ways to create a block:

1. Add a new block and drag stimuli into it

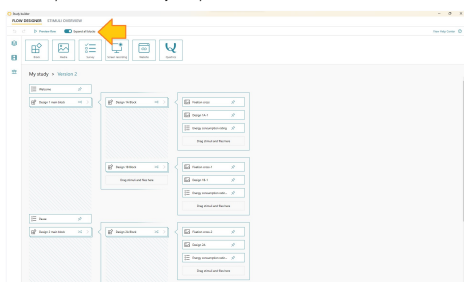


2. Multiselect stimuli and group them into a new block (Ctrl + G, or right-click context menu)



You can add as many layers of blocks as you need. Use the pinning and randomization function to decide in what order stimuli shall be shown inside a block, and in what order blocks shall be shown.

Tip: Use the **expand-all view** to review your experimental structure



The expand all view

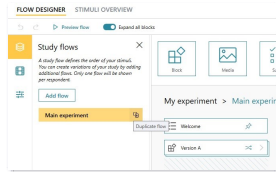
Working with multiple study flows

If you want to work between subject designs (A/B tests) and make parallel versions of your experiment, you can create multiple study flows.

This means you can add a new flow completely with new stimuli, reuse stimuli and blocks from an existing flow, or duplicate an existing flow for further modification.

Creating new flows

You can duplicate an existing flow or create an entirely new study flow.



Duplicate a flow

Working with duplicated study flows

Duplicating a flow means that you make a copy of a flow.

If a study flow was duplicated, the same stimuli are used in the flows derived from the original one. This means that

- They share the same name and
- That edits to stimulus properties (like presentation duration, manual advance, etc.) will be applied to all study flows where this stimulus is used.
- This also means that in the analysis, data from all flows can be aggregated onto the same stimulus.

While stimulus properties are shared across flows, the stimulus order (including the randomization property) is unique to a flow.

You can remove stimuli from a flow or any block in a flow without deleting this stimulus from other flows.

Making changes to the stimulus order within a block, or adding or removing (other) stimuli from a block does not affect other blocks with the same name in other study flows.

If you want to make unique changes to a stimulus (like having different presentation durations in different study flows) we recommend that you duplicate the stimulus.

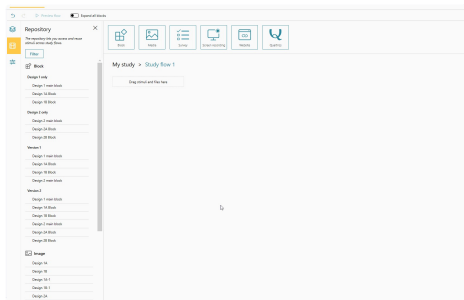
Working with the repository

The repository allows you to see which stimuli are used in the current study flow.

If you prefer to create a new flow from scratch as opposed to duplicating an existing one, the repository helps you to reuse stimuli from other flows.

Stimuli used in multiple flows share the name and all properties.

You can drag unused stimuli or blocks from the repository into your experimental structure.

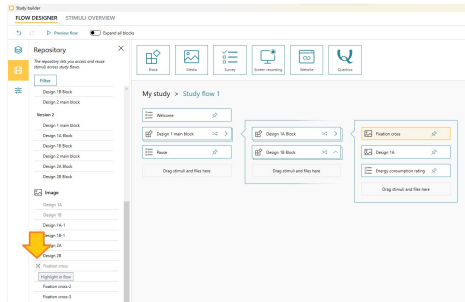


Please note that it is not possible to have the same stimulus twice in the same flow. iMotions will warn you in case you accidentally drag in the same stimulus twice.

Renaming a stimulus or changing its properties will change this stimulus on all other flows where it is used, too.

While stimulus properties are shared across flows, the stimulus order (including the randomization property) is unique to a flow.

You can locate a selected item in the flow by clicking the "Highlight in flow" icon



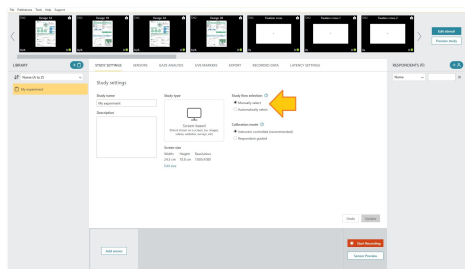
How to highlight a stimulus in a flow

Choosing which study flow to test on which respondents

If your study contains multiple study flows, you have to options to decide which flow to present to a respondent.

1. Choose that iMotions shall randomly pick a study flow when you start a new recording
2. Select the study flow manually after starting a new recording

You can set this choice in the study settings.



Configure the study flow selection

